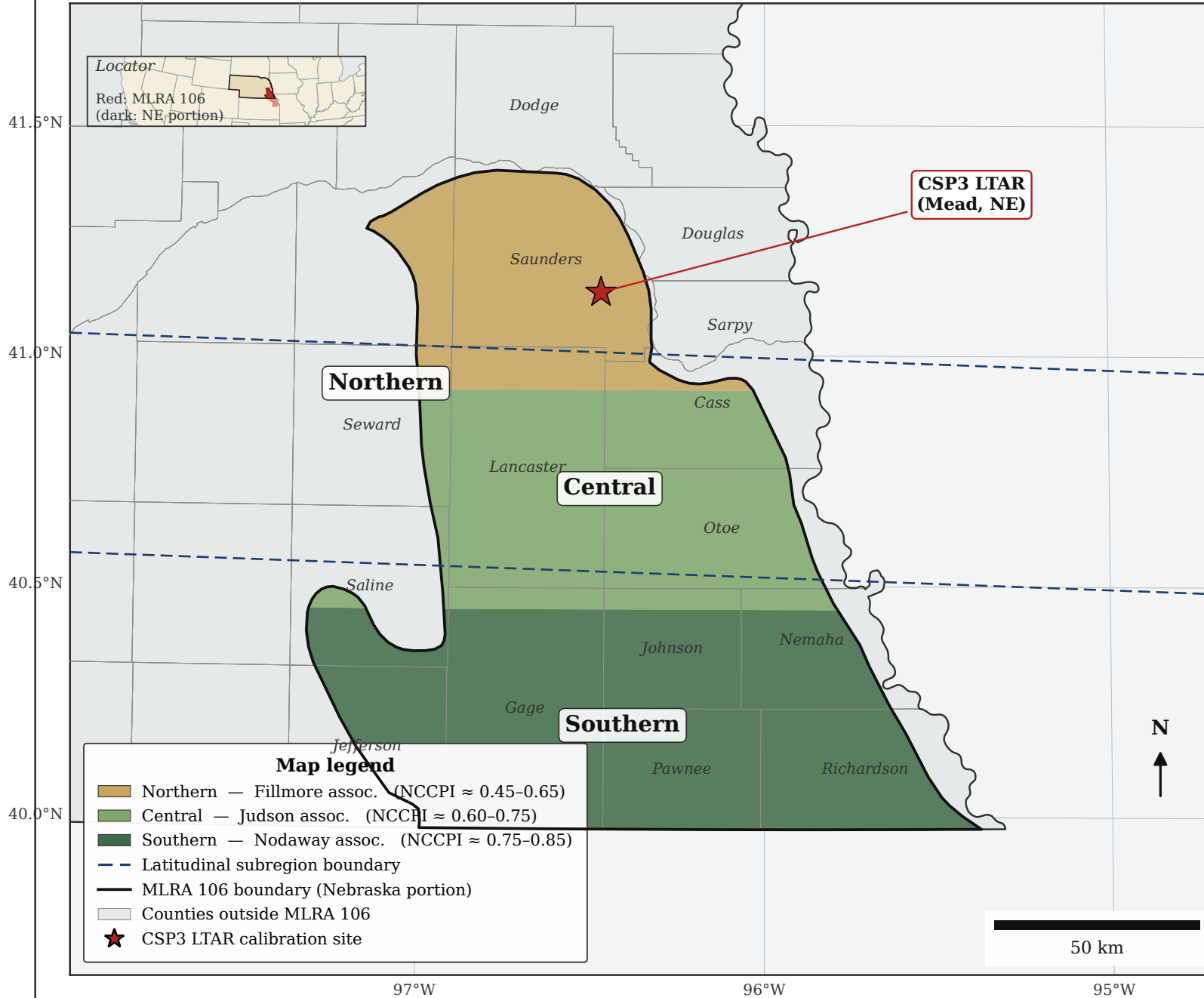


Figure 1. Major Land Resource Area (MLRA) 106 — Nebraska portion

Study-area extent, county boundaries, latitudinal subregions, CSP3 LTAR calibration site, and dominant NCCPI soil associations



Projection & datum

NAD83 / Conus Albers
Equal Area (EPSG:5070)
Standard parallels 29.5°N, 45.5°N
Latitude of origin 23°N

Data sources

- NRCS MLRA v5.2 (2022)
- TIGER/Line counties (2023)
- SSURGO dominant-component map units (Soil Survey Staff, 2020)
- USDA-ARS LTAR network (CSP3)
- National Commodity Crop Productivity Index (NCCPI v3.0)

Soil associations

Northern — Fillmore assoc.
NCCPI $\approx$ 0.45–0.65
Mollic Albaqualfs,
closed-depression
loess uplands

Central — Judson assoc.
NCCPI $\approx$ 0.60–0.75
Cumulic Hapludolls,
colluvial footslopes
on loess uplands

Southern — Nodaway assoc.
NCCPI $\approx$ 0.75–0.85
Mollic Udifluvents,
Holocene alluvium of
Missouri R. tributaries

Cartography

Equal-area statistics;
scale bar in projected
metres. Graticule at
1° / 0.5° spacing.

Prepared by the authors, 2026.
Reproducible: scripts/figure1.py